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tutorial 4

1. $G_h(s) = \dfrac{100}{(s+1)(s+2)}$

$G(j\omega) = \dfrac{100}{(j\omega+1)(j\omega+2)} = \dfrac{100}{-\omega^2+2+3j\omega}$

$|G(j\omega)| = \dfrac{100}{\sqrt{(-\omega^2+2)^2+(3\omega)^2}}$

$\angle G(j\omega) = \tan^{-1}\left(\dfrac{-3\omega}{-\omega^2+2}\right)$

$= -\tan^{-1}\left(\dfrac{3\omega}{-\omega^2+2}\right)$

Phase margin

$|G(j\omega)| = 1$

$\dfrac{100}{\sqrt{(-\omega^2+2)^2+(3\omega)^2}} = 1$

$100 = \sqrt{(-\omega^2+2)^2+(3\omega)^2}$

$9\omega^2 + \omega^4 - 4\omega^2 + 4 = 100^2$

$\omega^4 + 5\omega^2 + 4 - 100^2 = 0$

$\omega^2 = 97.5 \qquad \omega^2 = -102.51$

$\therefore \omega = 9.87$ rad/s.

when $\omega = 9.87$ rad/s

$\angle G(j\omega) = -\tan^{-1}\left(\dfrac{3\times 9.87}{-9.87^2+2}\right)$

$\boxed{\cancel{= -17.24 \text{ rad}} \quad \cancel{= 0.301 \text{ rad}}}$

$= -162.8^\circ$

Phase margin $= 180 - 162.8^\circ$

$= \underline{\underline{17.2^\circ}}$

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1) b)

i). from plot phase margin = 17°

ii) from plot gain crossover frequency = 10 rad/s

iii). if $PM = 45^\circ$

$$\angle G(j\omega) = -180 + 45$$
$$= -135^\circ$$

$$\angle G(j\omega) = -\tan^{-1}\omega - \tan^{-1}\frac{\omega}{2}$$

$$-135 = -\tan^{-1}\omega - \tan^{-1}\frac{\omega}{2}$$

$$135^\circ = +\tan^{-1}\omega + \tan^{-1}\frac{\omega}{2}$$

$$\tan(135) = \frac{\tan\omega + \frac{\omega}{2}}{1 - \frac{\omega^2}{2}}$$

$$-1 = \frac{\frac{\omega^2}{2}}{1 - \frac{\omega^2}{2}} \qquad -1 = \frac{\frac{3\omega}{2}}{1 - \frac{\omega^2}{2}}$$

$$\frac{\omega^2}{2} - 1 = \frac{\omega^2}{2} \qquad \frac{\omega^2}{2} - 1 = \frac{3\omega}{2}$$

$$\omega^2 - 3\omega - 2 = 0$$

$$\cancel{(\omega - 2)(\omega + 1)}$$

$$\omega = 3.56, \quad \omega = -0.56$$

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$\therefore \quad \omega = 3.56$

~~$|G(j\omega)H(j\omega)| = 1$~~

$|G(j\omega).k| = 1$

$$\frac{100k}{\sqrt{(2-\omega^2)^2 + 9\omega^2}} = 1$$

for $\omega = 3.56$.

$$\frac{100k}{} = \sqrt{(2-3.56^2)^2 + 9 \times 3.56^2}$$

$$\underline{\underline{k = 0.15}}$$

required gain $\underline{\underline{k = 0.15}}$

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2). i)

a) $G(s) = \dfrac{1}{s(s+2)(s+4)}$

$G(j\omega) = \dfrac{1}{j\omega(j\omega+2)(j\omega+4)}$

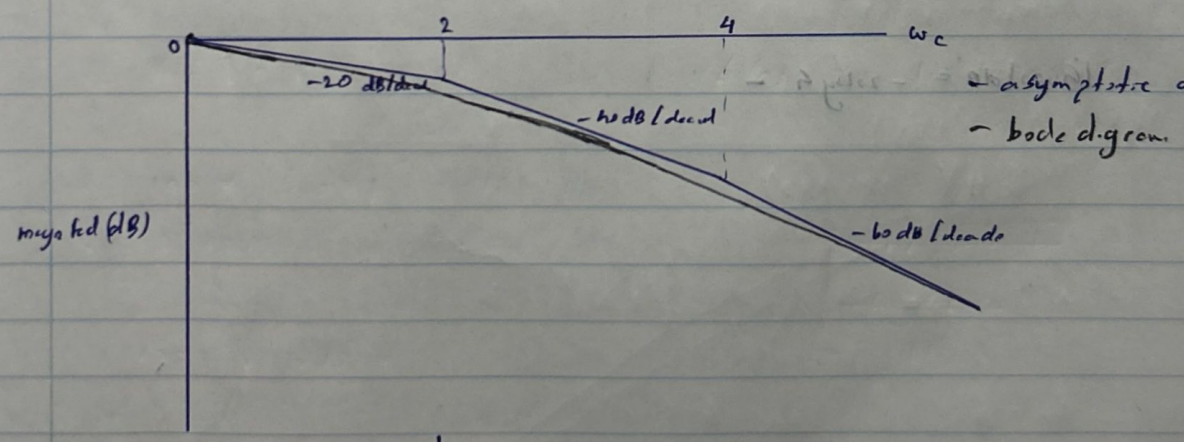
$|G(j\omega)| = \dfrac{1}{\omega\sqrt{(\omega^2+4)(\omega^2+16)}}$, $\angle G(j\omega) = -90 - \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}\left(\frac{\omega}{4}\right)$.

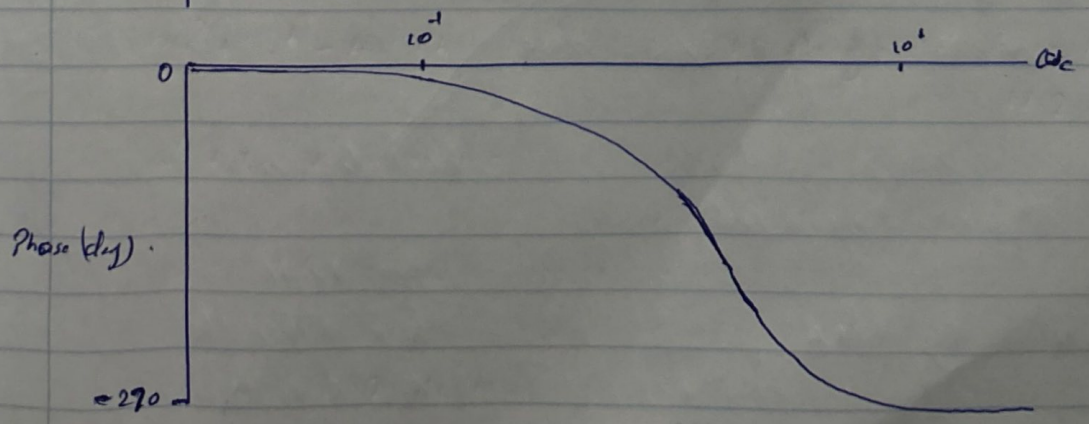
$|G(j\omega)|_{dB} = -20\log\omega - 10\log(\omega^2+4) - 10\log(\omega^2+16)$

When $\omega < 2$ $\quad |G(j\omega)|_{dB} = -20\log\omega$ $\quad \angle G(j\omega) \approx 0$ for small ω.

$2 < \omega < 4$ $\quad |G(j\omega)|_{dB} = -40\log\omega$. $\quad \angle G(j\omega) = -220$ for large ω.

$\omega > 4$ $\quad |G(j\omega)| = -60\log\omega$.





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b).

$$G(s) = \frac{1}{(s+1)(s+2-j)(s+2+j)}$$

$$= \frac{1}{(s+1)(s^2+4s+5)}$$

$$G(j\omega) = \frac{1}{(j\omega+1)(-\omega^2+j4\omega+5)}$$

$$G(j\omega) = \frac{1}{(1+j\omega)((5-\omega^2)+j4\omega)}$$

$$s^2+4s+5 = s^2+2\zeta\omega_n s+\omega_n^2$$

$\omega_n^2 = 5$, $2\zeta\omega_n = 4$

$\omega_n = \sqrt{5}$ $\zeta = \frac{2}{\sqrt{5}} = 0.894$

break frequencies

$\omega = 1$ rad/s

$\omega = 2.236$ rad/s

$$|G(j\omega)|_{dB} = +20\log\left(\frac{1}{\sqrt{(1+\omega^2)((5-\omega^2)^2+(4\omega)^2)}}\right)$$

when ω small $|G(j\omega)|_{dB} = +20\log\frac{1}{1\sqrt{5^2+4^2}}$

$$= +20\log\frac{1}{5} = -13.98$$

$\angle G(j\omega) = -\tan^{-1}(\omega) - \tan^{-1}\left(\frac{4\omega}{5-\omega^2}\right)$

when ω small

$\angle G(j\omega) = 0$

at very high ω

$\frac{1}{s+1} \longrightarrow -90^\circ$

$\frac{1}{s^2+4s+5} \longrightarrow -180^\circ$

$\therefore \angle G(j\omega) = -270^\circ$

$\omega \ll 1$ -20 dB/decade slope.

$1 < \omega < \sqrt{5}$ -20 dB/decade slope.

$\omega > \sqrt{5}$ -40 dB/decade slope.

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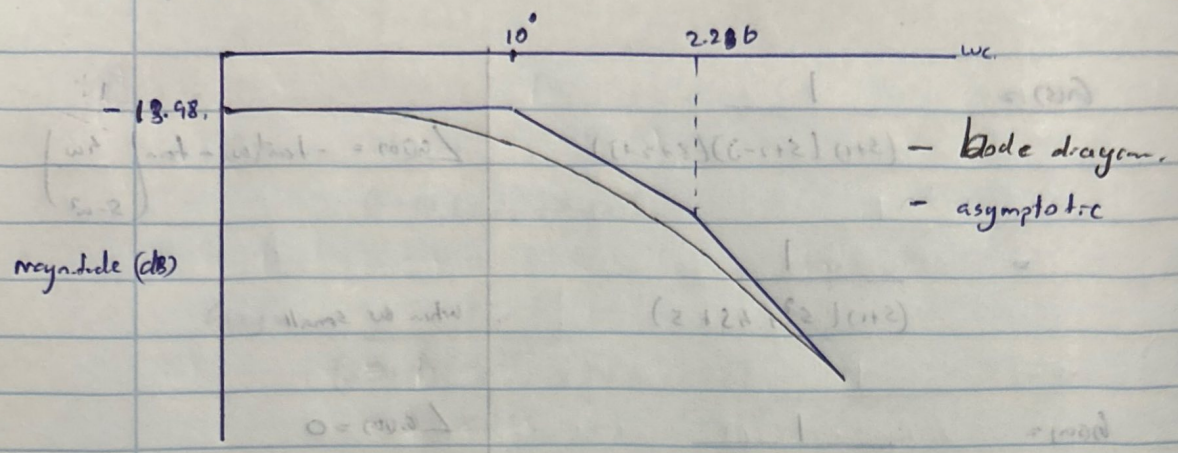

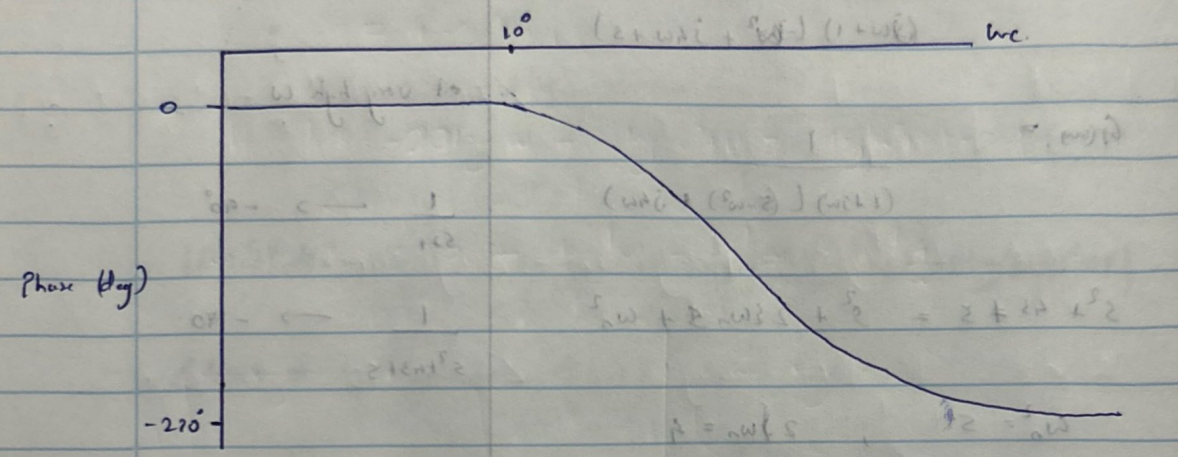

c). $G(s) = \dfrac{s+2}{(s+1)(s+3-j)(s+3j)}$

$G(s) = \dfrac{s+2}{(s+1)(s^2+6s+10)}$

$s^2 + 6s + 10 = s^2 + 2\xi\omega_n + \omega_n^2$

$\omega_n = \sqrt{10}$, $\quad 2\xi\omega_n = 6$

$\omega_n = 3.16 \quad \xi = 3/\sqrt{10} = 0.95$

$G(j\omega) = \dfrac{j\omega + 2}{(j\omega+1)(-\omega^2 + 10 + 6j\omega)}$

break frequencies

$\omega = 1$ rad/s

$\omega = 3.16$ rad/s

$\omega = 2$ rad/s

$|G(j\omega)|\,dB = 20\log \dfrac{\sqrt{\omega^2+4}}{\sqrt{(\omega^2+1)((10-\omega^2)^2 + 36\omega^2)}}$

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when ω is very small

$(G(j\omega))_{dB} = 20\log 4$

$= 12.04$ dB.

$\omega < 1$	0 db/decade
$1 < \omega < 2$	-20 db/decade
$2 < \omega < 3.16$	0 db/decade.
$\omega > 3.16$	-20 db/decade.

$\angle G(j\omega) = \tan^{-1}\frac{\omega}{2} - \tan^{-1}\omega - \tan^{-1}\left(\frac{6\omega}{10-\omega^2}\right)$

ω is very small

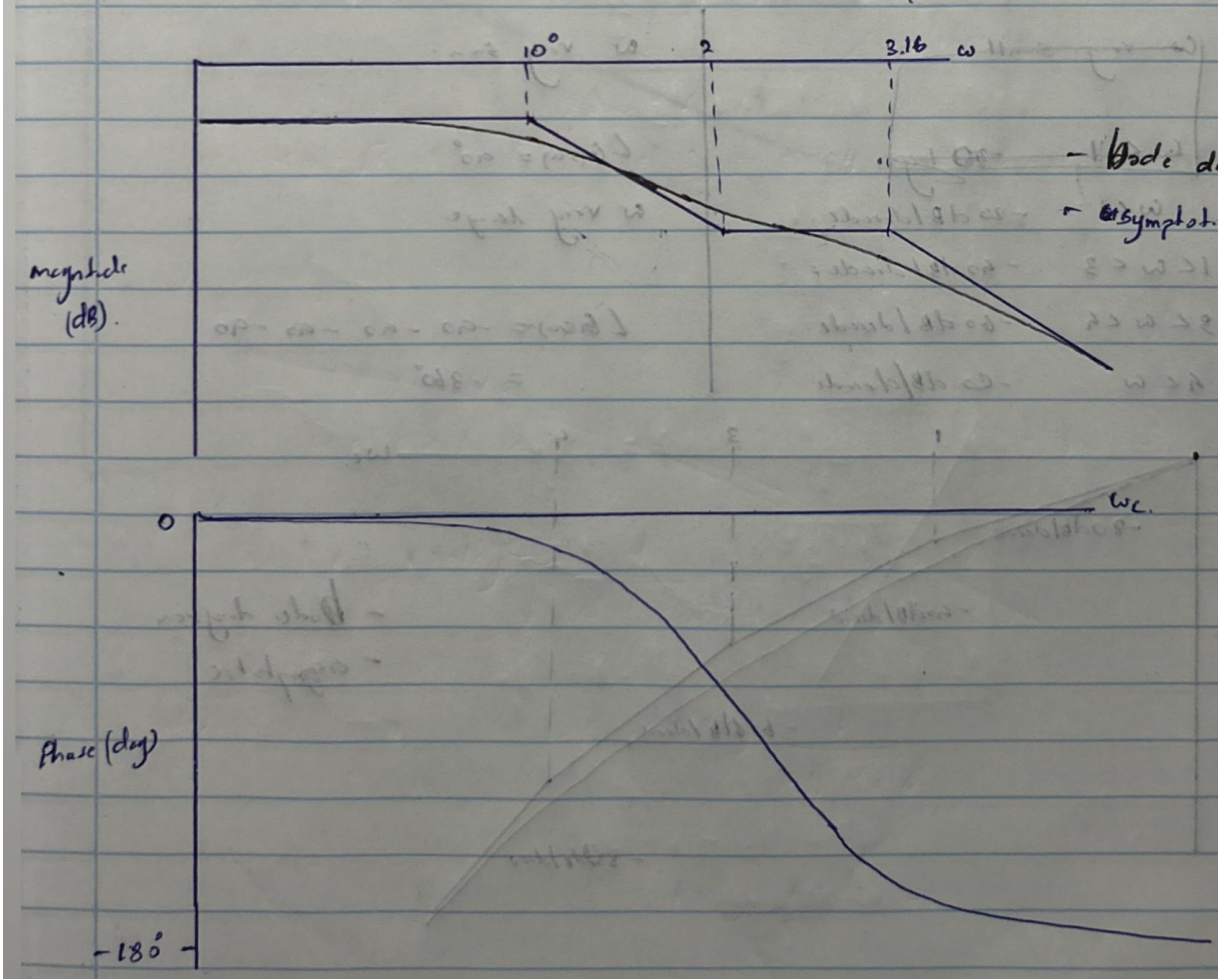
$\angle G(j\omega) = 0$

$\omega \to \infty$ $\tan^{-1}\frac{\omega}{2} \to 90^\circ$

$-\tan^{-1}\omega \to -90$

$-\tan^{-1}\left(\frac{6\omega}{10-\omega^2}\right) \to -180^\circ$

$\therefore \angle G(j\omega) \to -180^\circ$



- Bode diagram
- asymptotic.

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d). $G(s) = \dfrac{1}{s(s+1)(s+3)(s+4)}$.

break frequencies

$\omega = 1$ rad/s

$\omega = 3$ rad/s

$\omega = 4$ rad/s

$|G(j\omega)|_{dB} = -20\log\omega - 20\log(\omega^2+1) - 20\log(\omega^2+9) - 20\log(\omega^2+16)$

$\angle G(j\omega) = -90 + \tan^{-1}\omega - \tan^{-1}\frac{\omega}{3} - \tan^{-1}\frac{4}{\omega}$

~~ω very small~~

~~ω < 1 -20 log~~

$\omega < 1$ -20 dB/decade.

$1 < \omega < 3$ -40 dB/decade.

$3 < \omega < 4$ -60 dB/decade.

$4 < \omega$ -80 dB/decade

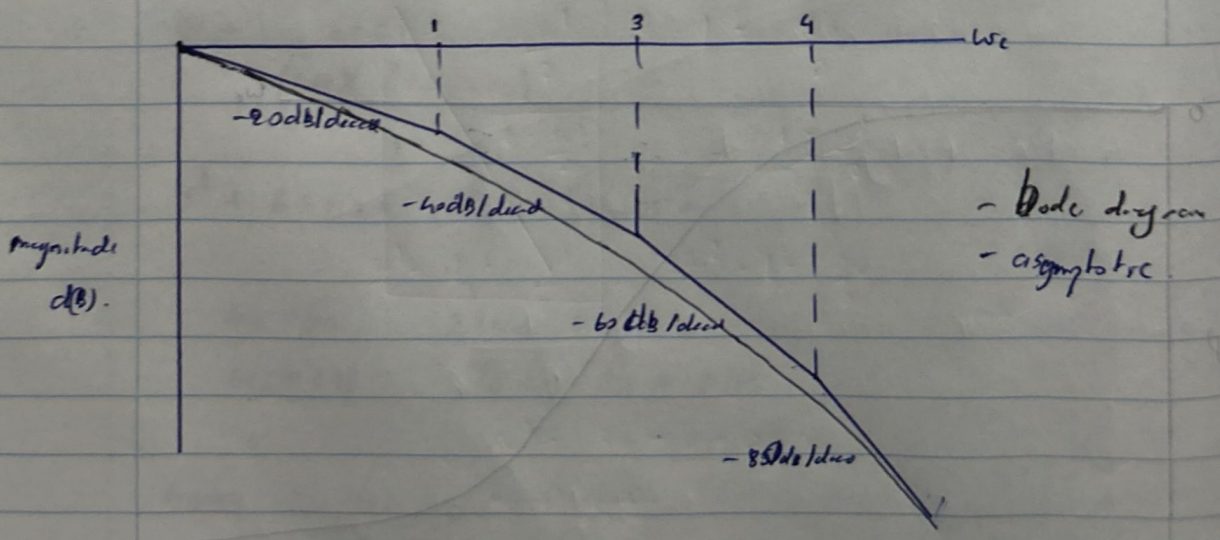
ω very small.

$\angle G(j\omega) = 90°$

ω very large

$\angle G(j\omega) = -90 - 90 - 90 - 90$

$= -360°$



- bode diagram
- asymptotic

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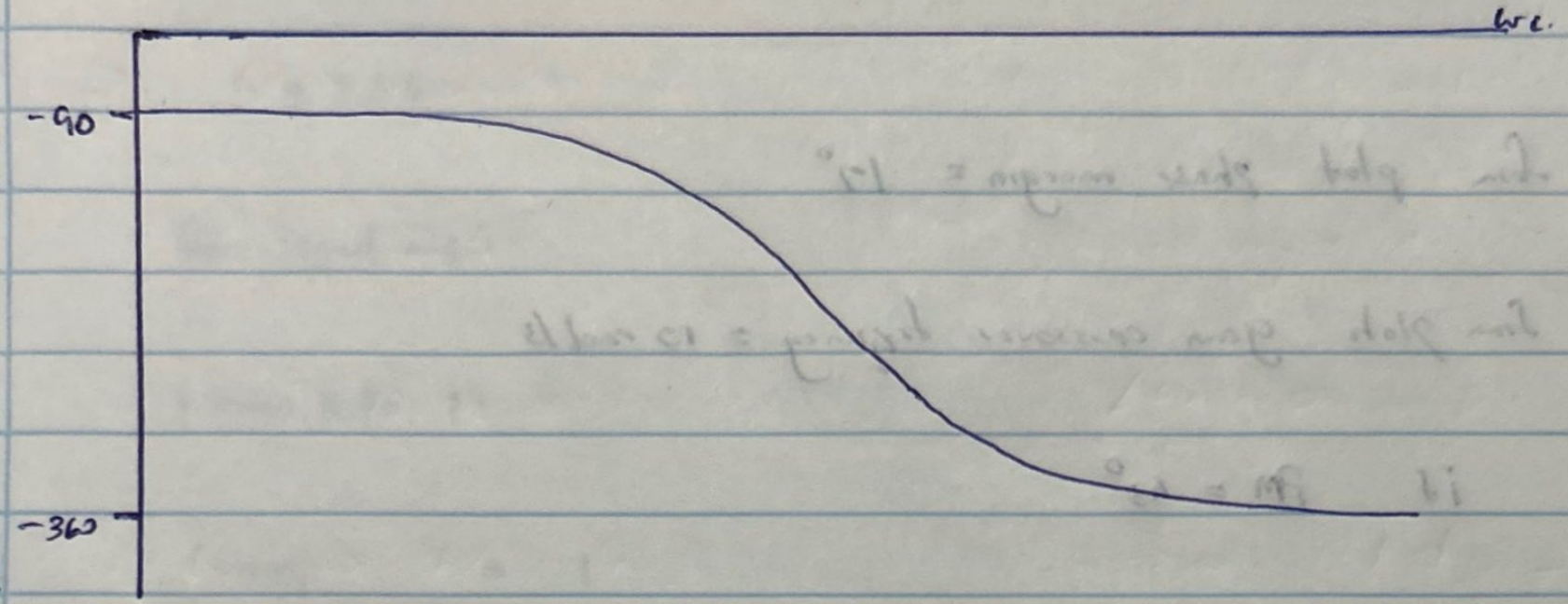
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